

One-Day-Ahead 1% VaR Forecasting on QQQ Daily Log Returns

A stepwise model-selection study

Ethan Yao

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1 Introduction

Value at risk is the maximum loss level expected not to be exceeded except with small probability α over a chosen time horizon. Financially, VaR is used to summarize portfolio downside risk in one number for trading, risk management, and capital allocation.

This report studies one-day-ahead 1% Value-at-Risk (VaR) forecasting for QQQ daily log returns. I will start with simple volatility-based benchmarks, identify their weaknesses, and use those weaknesses to motivate the next improvement.

The model path is therefore:

- RiskMetrics ($\lambda = 0.94$) and Gaussian GARCH(1,1) as baseline volatility-VaR models,
- Student- t GARCH(1,1) once Gaussian tails prove inadequate,
- two extension paths: richer GARCH-family specifications and alternative nonparametric or direct-quantile approaches,
- and finally empirical residual tail calibration, which leads to the strongest results.

2 Executive Summary

The research question is not volatility forecasting alone. For 1% left-tail risk, the key issue is tail calibration. A model can track conditional variance reasonably well and still underestimate downside risk if its lower tail is too thin. The evidence in this study shows exactly that pattern. Gaussian baselines are too shallow, Student- t innovations help, asymmetric volatility helps somewhat, but the largest improvement comes from empirical-tail calibration through filtered historical simulation (FHS) and related bootstrap methods.

Over the common out-of-sample period of 848 trading days, GJR-GARCH with empirical tail calibration (implemented here through Filtered Historical Simulation) emerges as the preferred model, with GARCH-FHS a close second. Historical Simulation remains a credible benchmark, as its simplicity does not prevent it from delivering competitive results. GARCHX(VIX), however, is retained only as a comparison benchmark, because the inclusion of VIX as an exogenous regressor is not supported by the final performance evidence in this project. The simplified CAViaR-SAV specification is interesting from a methodological perspective, but its results are not strong enough to displace the final recommendation.

3 Historical Background

VaR became especially prominent in the 1990s as institutions sought a compact summary of downside market risk. J.P. Morgan's RiskMetrics framework helped standardize daily variance-based VaR forecasting, which is why it remains a natural benchmark.

The econometric basis of modern VaR forecasting comes from conditional heteroskedasticity models. Engle's ARCH model and Bollerslev's GARCH extension formalized volatility clustering and provided a natural route from conditional variance forecasts to conditional quantile forecasts. Once a model for time-varying volatility is available, one-step-ahead VaR can be constructed from a conditional mean, a conditional standard deviation, and an assumed or estimated lower-tail quantile.

Evaluation then became as important as construction. Kupiec (1995) proposed unconditional coverage testing, while Christoffersen (1998) showed that correct hit frequency is not enough if violations cluster through time.

A second major development was the move away from purely Gaussian tails. Financial returns often exhibit heavier tails than Gaussian theory implies, which motivated Student- t innovations and semi-parametric methods. Barone-Adesi, Giannopoulos, and Vosper (1999) proposed Filtered Historical Simulation, which combines a volatility model with the empirical distribution of standardized residuals. That idea is central here because the strongest improvement in this report comes from empirical-tail calibration.

A different strand of the literature models the conditional quantile directly. The leading example is CAViaR (Engle and Manganelli, 2004). It is attractive because VaR itself is modeled directly, but in practice it requires careful optimization and adequacy testing.

4 Purpose Statement and Mathematical Development

4.1 VaR and the forecasting objective

Let r_t denote the QQQ daily log return,

$$r_t = \log \left(\frac{P_t}{P_{t-1}} \right).$$

For tail probability $\alpha = 0.01$, the one-day-ahead VaR is the conditional lower-tail quantile satisfying

$$\Pr(r_{t+1} < \text{VaR}_{t+1}(0.01) \mid \mathcal{F}_t) = 0.01.$$

For many models in this report,

$$r_{t+1} = \mu_{t+1} + \sigma_{t+1} z_{t+1},$$

so the forecast takes the form

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_\alpha.$$

This decomposition is useful because the models differ in two main ways:

1. how they forecast σ_{t+1} ,
2. how they obtain the tail quantile $\hat{q}_\alpha$.

The stepwise comparison in Section 5 is therefore a sequence of attempts to improve one or both parts of the VaR forecast.

4.2 Rolling out-of-sample design

The empirical study uses:

- 2,824 cleaned observations,
- a 70% initial training split,
- 848 one-step-ahead out-of-sample forecasts,
- an expected number of 1% violations equal to 8.48.

At each forecast origin, the model is re-estimated using information available up to that date and then used to forecast next-day VaR. This rolling design mirrors how VaR would be used in practice.

4.3 Violation indicator

Given a forecast $\widehat{\text{VaR}}_t$, define the hit indicator

$$I_t = \mathbf{1}\{r_t < \widehat{\text{VaR}}_t\}.$$

If the model is correctly calibrated at the 1% level, the sequence I_t should have mean close to 0.01 and should not exhibit systematic clustering.

4.4 Data Example

The study uses QQQ adjusted closing prices to construct daily log returns and uses VIX closing values only in the GARCHX benchmark. A representative forecast date is 2025-04-04, when QQQ experienced a large negative return. (this is a single stressed example, not to rank models on the basis of one date)

Date	Model	VaR	Return	Viol.
2025-04-04	GJR-FHS	-0.0796	-0.0641	No
2025-04-04	GARCH-FHS	-0.0706	-0.0641	No
2025-04-04	Student- t GARCH	-0.0631	-0.0641	Yes
2025-04-04	Historical Simulation	-0.0366	-0.0641	Yes
2025-04-04	RiskMetrics(0.94)	-0.0453	-0.0641	Yes
2025-04-04	GARCHX(VIX)	-0.0491	-0.0641	Yes
2025-04-04	CAViaR-SAV	-0.0629	-0.0641	Yes

The example shows the broad pattern seen in the full backtest. During stress, the empirical-tail models move substantially further into the left tail, while several baseline models remain too close to zero and are breached. Representative implementation snippets are collected in **Appendix**.

4.5 Backtesting and Forecast Evaluation Metrics

The comparison uses four evaluation tools.

4.5.1 Kupiec unconditional coverage test

The null hypothesis is that the violation probability equals the target level α . If x is the number of violations in n forecasts and $\hat{\pi} = x/n$, the likelihood-ratio statistic is

$$LR_{uc} = -2 \log \left[\frac{(1 - \alpha)^{n-x} \alpha^x}{(1 - \hat{\pi})^{n-x} \hat{\pi}^x} \right].$$

A large p-value means the observed hit rate is plausible under the target 1% violation probability.

4.5.2 Christoffersen independence test

Correct hit frequency is not enough if violations cluster. The independence test compares an independent-hit model with a first-order Markov alternative. A low p-value indicates that violations are arriving in a dependent pattern.

4.5.3 Conditional coverage test

The conditional coverage test combines the previous two ideas:

$$LR_{cc} = LR_{uc} + LR_{ind}.$$

A good VaR model should perform reasonably well on both dimensions.

4.5.4 Check Loss

Because violation-based backtests ignore the size of forecast errors, the report also uses quantile check loss,

$$\rho_\alpha(u) = u(\alpha - I(u < 0)), \quad u_t = r_t - \widehat{\text{VaR}}_t.$$

Equivalently,

$$\text{CheckLoss}_t = (r_t - \widehat{\text{VaR}}_t) (\alpha - I(r_t < \widehat{\text{VaR}}_t)).$$

For non-violation days the penalty is small but nonzero, so overly conservative VaR still incurs loss. For violation days the penalty is much larger, especially when the realized loss falls far below the forecast threshold.

The model-level values reported throughout this report are mean check losses, i.e. the average of the 848 pointwise check-loss values for each model.

5 Model Development and Comparison

5.1 Baseline Gaussian volatility models

The natural starting point is the standard volatility-to-VaR pipeline used in many introductory treatments: forecast conditional variance, then map that forecast into VaR with a Gaussian quantile. The two baseline models are **RiskMetrics** ($\lambda = 0.94$) and **Gaussian GARCH(1,1)**.

RiskMetrics uses the EWMA recursion

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2, \quad \lambda = 0.94,$$

with one-step forecast

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \Phi^{-1}(\alpha).$$

Gaussian GARCH(1,1) writes

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \sim N(0, 1),$$

with variance recursion

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2.$$

Model	Viol.	Rate	Kupiec p	Christ. p	CC p	Mean Check Loss
RiskMetrics	18	2.12%	0.0043	0.3925	0.0117	0.0004458
GARCH- Gaussian	22	2.59%	0.0001	0.5970	0.0005	0.0004363

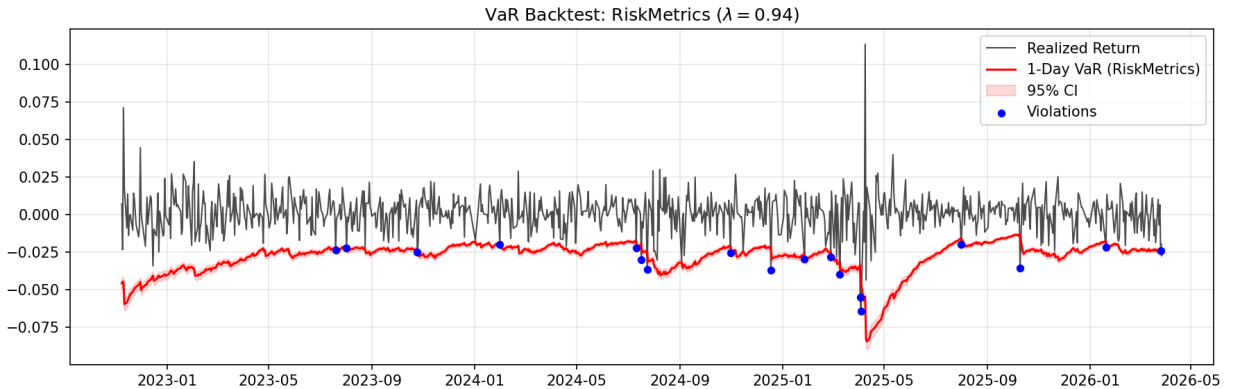


Figure 1: RiskMetrics ($\lambda = 0.94$) backtest

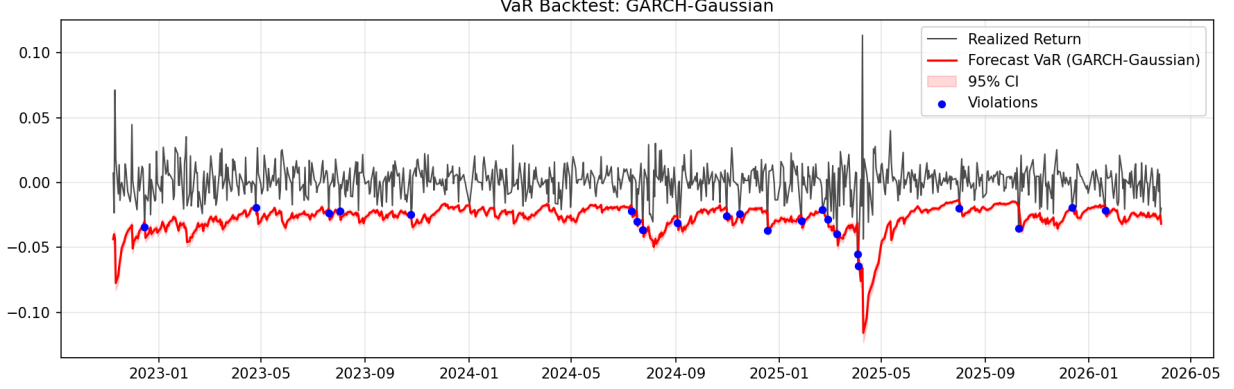


Figure 2: Gaussian GARCH(1,1) backtest

Both benchmark models underestimate downside risk. RiskMetrics is smoother and more persistent, while GARCH-Gaussian reacts a bit faster and becomes more negative in stress, but neither model pushes the VaR line far enough into the left tail on a consistent basis.

That visual impression is backed up by the backtests: RiskMetrics records 18 violations and GARCH-Gaussian records 22, versus only 8.48 expected at the 1% level; their conditional coverage p-values are 0.0117 and 0.0005, respectively, so both fail as baseline calibrations.

5.2 Heavier-tailed parametric improvement: Student- t GARCH

Once the Gaussian baselines underperform, the next step is to keep the GARCH volatility structure but allow heavier-tailed innovations.

Student- t GARCH uses the same variance recursion as Gaussian GARCH, but the random innovation z_t follows a standardized Student- t distribution with ν degrees of freedom. $\hat{q}_\alpha^{(t)}$ is the fitted standardized Student- t 1% quantile, the VaR forecast is:

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_\alpha^{(t)},$$

Model	Viol.	Rate	Kupiec p	Christ. p	CC p	Mean Check Loss
GARCH(1,1) Student- t	13	1.53%	0.1480	0.1890	0.1482	0.0004131

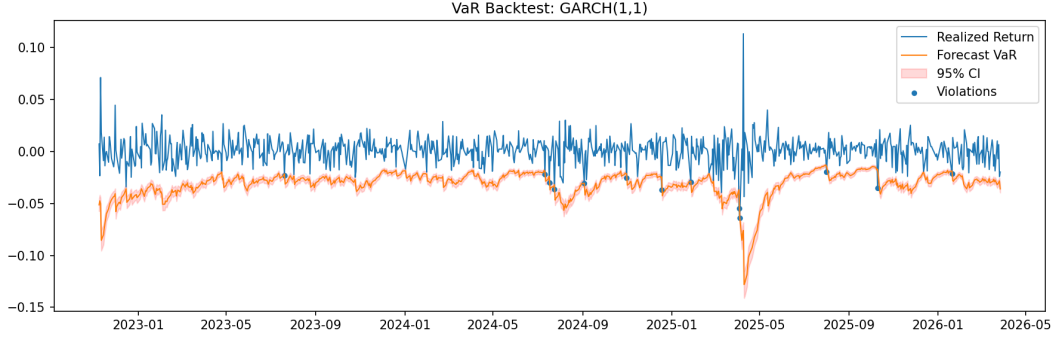


Figure 3: Student- t GARCH(1,1) backtest

The VaR line moves deeper into the left tail during volatile periods. Even so, it is still breached too often in stress clusters. The backtests show that Student- t innovations help, but do not fully solve the calibration problem.

5.3 Extension path 1: GARCH-family extensions

Two richer volatility specifications are considered before trying alternative nonparametric or direct-quantile approaches.

5.3.1 GARCHX(VIX)

GARCHX(VIX) adds an exogenous volatility regressor to the variance equation,

$$\begin{aligned}\varepsilon_t &= r_t - \mu \\ \varepsilon_t &= \sigma_t W_t, W_t \sim N(0, 1) \\ \sigma_{t+1|t}^2 &= \omega + \alpha \varepsilon_t^2 + \beta \sigma_t^2 + \delta x_t\end{aligned}$$

where x_t is standardized VIX. Economically, this is plausible because VIX summarizes market-implied uncertainty. Empirically, however, it is weaker than the leading models in this report.

Model	Viol.	Rate	Kupiec p	Christ. p	CC p	Mean Check Loss
GARCHX(VIX)	15	1.77%	0.0424	0.2621	0.0679	0.0004185

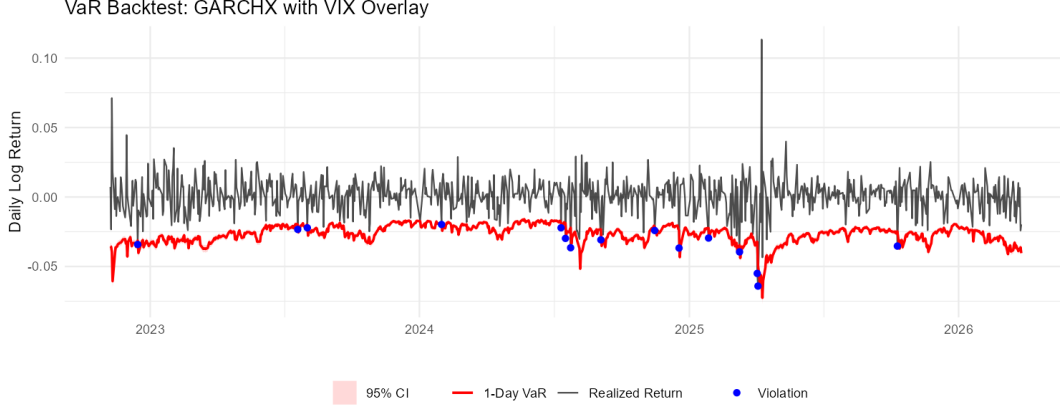


Figure 4: GARCHX(VIX) backtest

GARCHX with VIX does react to rising market stress, but in this plot the improvement is limited. The VaR line still stays too shallow in the main stress episodes, and we still see several clustered breaches. The backtest p-values are only better than the Gaussian baselines, but not strong enough to make GARCHX a leading recommendation.

5.3.2 GJR-GARCH

A more promising extension is asymmetric volatility. GJR-GARCH uses

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 I(\varepsilon_{t-1} < 0) + \beta \sigma_{t-1}^2.$$

If $\gamma > 0$, negative shocks increase future volatility more strongly than positive shocks. This is plausible because in financial markets, negative returns are more closely tied to market stress, so the left tail should react more strongly to bad news.

Model	Viol.	Rate	Kupiec p	Christ. p	CC p	Mean Check Loss
GJR-GARCH	15	1.77%	0.0424	0.4621	0.0972	0.0004069

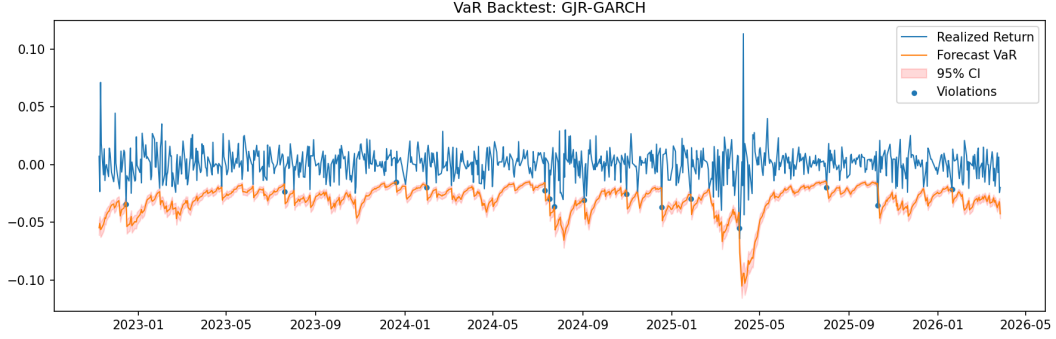


Figure 5: GJR-GARCH backtest plot.

The plot shows that GJR-GARCH reacts coherently to changing volatility, but it still doesn't push the VaR line far enough into the left tail during stress. GJR-GARCH responds somewhat better to negative shocks compared to GARCHX, which is why it is the more promising extension, but asymmetry alone is still not enough.

5.4 Extension path 2: Nonparametric and Direct quantile modeling

Rather than modeling volatility first and then mapping it to VaR, quantile forecast models estimate the tail threshold directly.

5.4.1 Historical Simulation

Historical Simulation is a deliberately simple nonparametric benchmark. Using a rolling window of size $W = 250$,

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \text{Quantile}_{\alpha}(r_{t-W+1}, \dots, r_t).$$

Model	Viol.	Rate	Kupiec p	Christ. p	CC p	Mean Check Loss
Historical Simulation	12	1.42%	0.2528	0.1569	0.1910	0.0004826

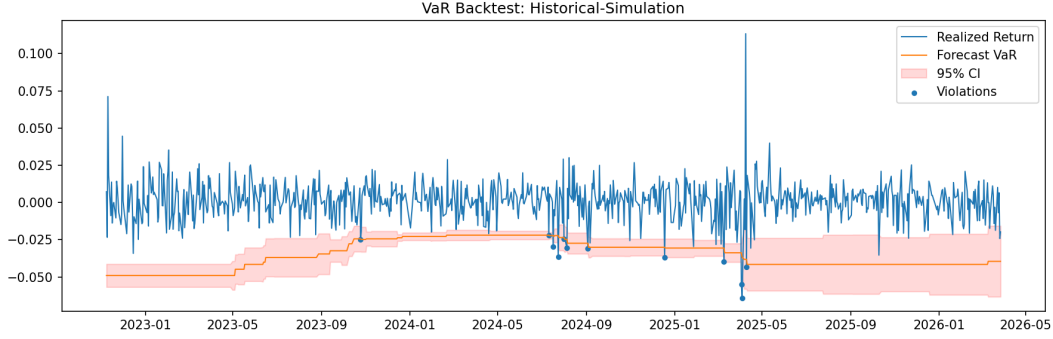


Figure 6: Historical Simulation backtest plot.

The Historical Simulation plot shows why this model remains a credible benchmark despite its simplicity. Its VaR path adjusts in a visibly stepwise way and can lag after volatility regime changes, but it still tracks stressed periods well enough to stay competitive in the backtests.

One clear weakness is that it misses the sharp volatility spike, and its confidence interval is visibly wider than those of the other models.

5.4.2 CAViaR-SAV

CAViaR-SAV provides an autoregressive specification for the VaR process, updating the quantile through its own lag and the magnitude of the most recent return. In this study, it uses a simplified SAV specification motivated by Engle and Manganelli’s CAViaR framework.

$$v_t = \beta_0 + \beta_1 v_{t-1} + \beta_2 |r_{t-1}|,$$

and the left-tail quantile is recovered through $q_t = -v_t$, where r_t is the realized return and v_t is the positive latent tail-risk level that the CAViaR recursion updates.

Parameters are estimated by minimizing the quantile check-loss objective,

$$L(\beta) = \sum_t \rho_\alpha(r_t - q_t(\beta)), \quad \rho_\alpha(u) = u(\alpha - I(u < 0)).$$

The implementation in this study is simplified. Only the SAV form is included, optimization is relatively light, and the Dynamic Quantile adequacy test from the original CAViaR framework is not part of this study.

Model	Viol.	Rate	Kupiec p	Christ. p	CC p	Mean Check Loss
CAViaR-SAV	6	0.71%	0.3664	0.0297	0.0625	0.0004010

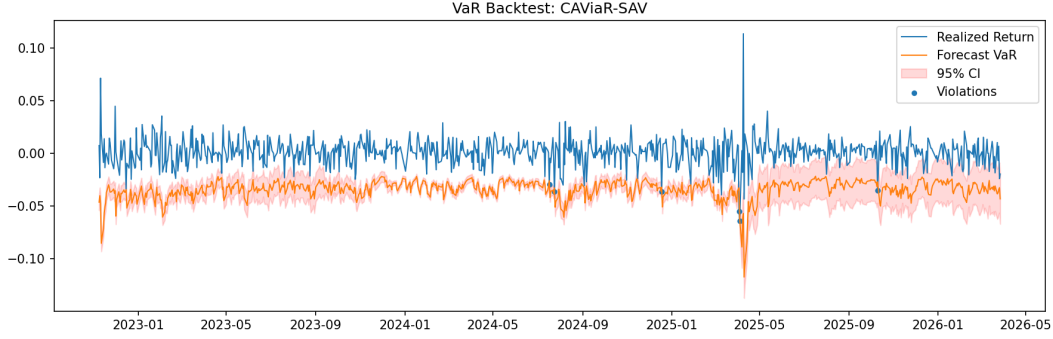


Figure 7: CAViaR-SAV backtest plot.

The CAViaR-SAV plot shows why this result is mixed rather than clearly successful. The quantile path often sits at a plausible level, 6 violations versus 8.48 expected is acceptable hit count. But the exceptions are not well spread through time, and that is the main weakness.

The backtests confirm this pattern. Coverage is acceptable, while the independence result is weak, so the model gets the overall level roughly right but not the timing of tail events.

The confidence band also widens sharply after the volatility spike because the interval is built from a shifted recent return window. Once that window includes the spike, empirical tail uncertainty rises and the band widens. Overall, the model looks slightly conservative rather than clearly well calibrated, and the plot supports the same conclusion as the independence test: the level can look reasonable even when the dynamic exception pattern is not convincing.

5.5 Non-parametric Quantile-GARCH: empirical-tail FHS / bootstrap methods

By this stage the pattern is clear. Gaussian tails are too thin, Student- t helps, asymmetry helps somewhat, Historical Simulation is a serious benchmark, and CAViaR-SAV is intriguing but not fully convincing. The strongest remaining step is to keep the volatility model but estimate the lower tail empirically rather than through a fixed parametric law.

5.5.1 Empirical tails with plain GARCH

After fitting a volatility model, standardized residuals are computed as

$$\hat{z}_t = \frac{r_t - \hat{\mu}_t}{\hat{\sigma}_t},$$

and the one-step forecast uses the empirical 1% quantile of those residuals,

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_{\alpha, \text{emp}},$$

where $\hat{q}_{\alpha, \text{emp}} = \hat{F}^{-1}(\alpha)$ is the empirical quantile of GARCH residuals.

On the other hand, Bootstrap fits GARCH first, then resamples the fitted standardized residuals, simulates many one-step returns,

$$r_{t+1}^{*(i)} = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} z_{t+1}^{*(i)},$$

and takes the empirical α -quantile of those simulated returns as VaR.

For one-step forecasting, bootstrap is essentially just a simulation-based way to recover the same empirical residual tail used by FHS. That is why, in this study, bootstrap and FHS give very similar results.

Model	Viol.	Rate	Kupiec p	Christ. p	CC p	Mean Check Loss
GARCH-FHS	7	0.83%	0.5984	0.7327	0.8212	0.0004126
GARCH- t -FHS	6	0.71%	0.3664	0.7698	0.6371	0.0004258
GARCH-Bootstrap	5	0.59%	0.1934	0.8075	0.4166	0.0004154
GARCH- t -Bootstrap	6	0.71%	0.3664	0.7698	0.6371	0.0004302

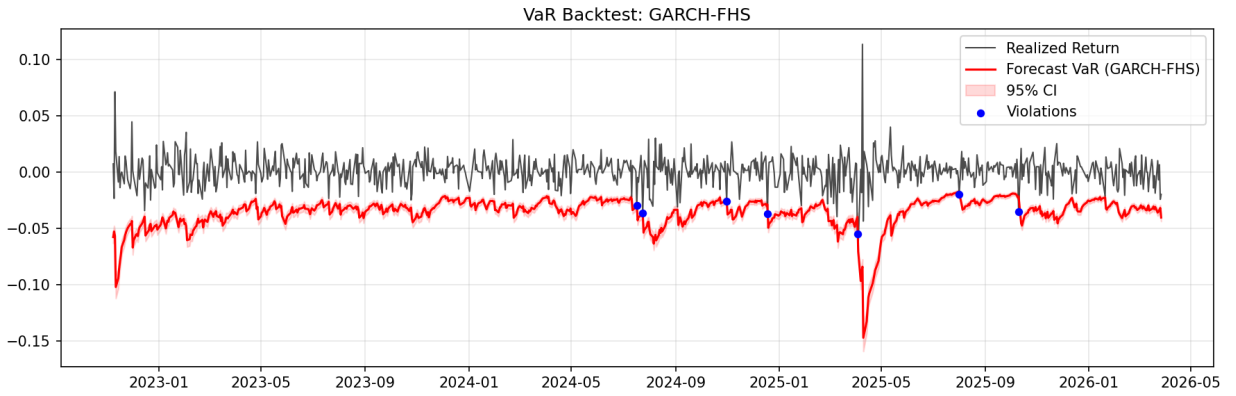


Figure 8: GARCH-FHS backtest plot.

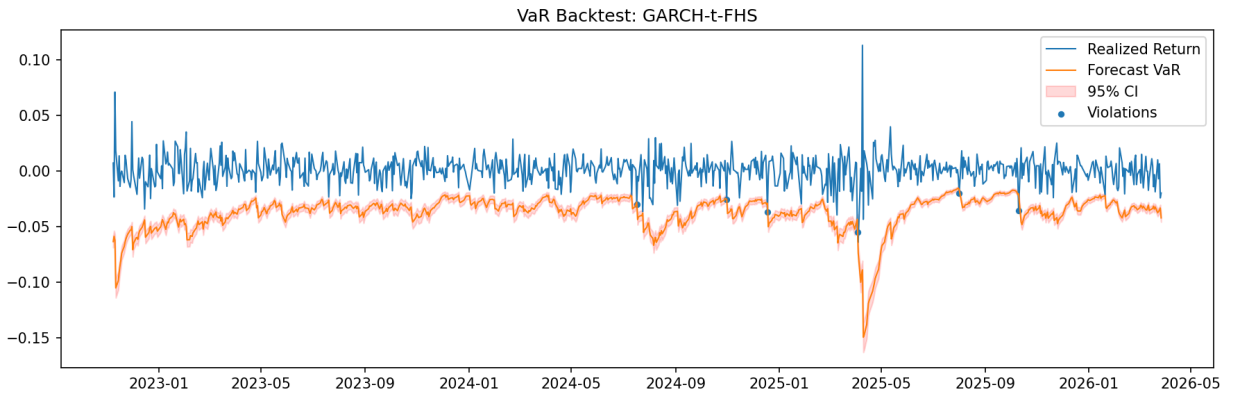


Figure 9: GARCH- t -FHS backtest plot.

The paired FHS plots show the main visual effect of empirical-tail calibration: both models move further into the left tail during stress than the purely parametric versions. The gain is not only in depth but also in timing, since the VaR line adapts more convincingly when volatility rises sharply.

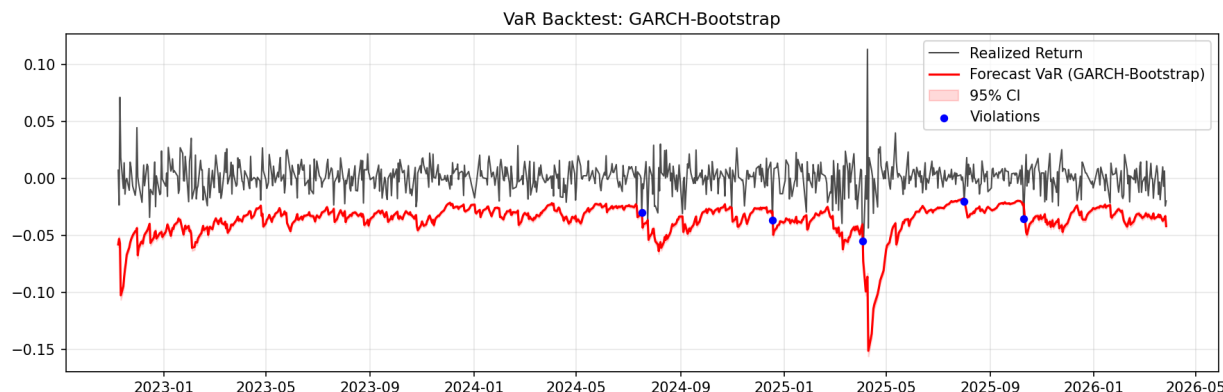


Figure 10: GARCH-Bootstrap backtest plot.

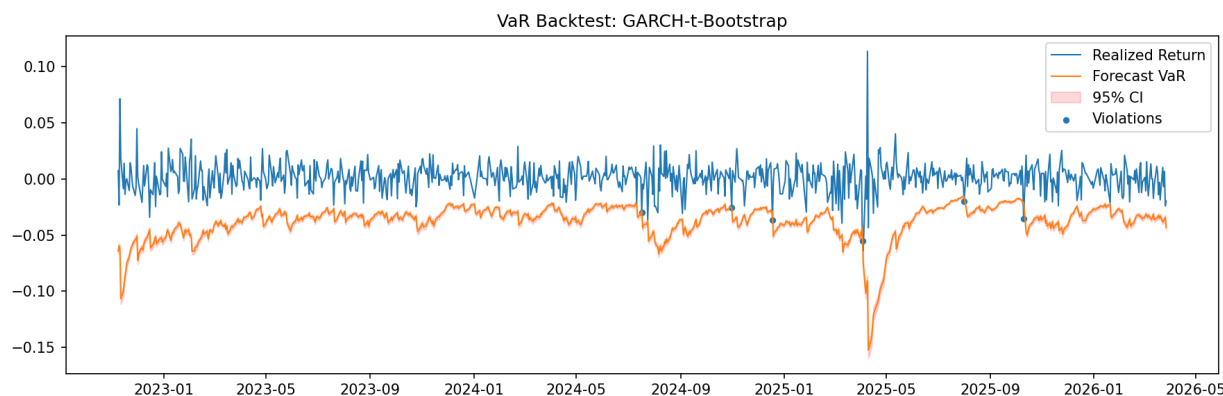


Figure 11: GARCH- t -Bootstrap backtest plot.

The paired bootstrap plots look very close to the FHS plots, which is exactly what one would expect in one-step forecasting when both methods rely on empirical standardized residual tails. The bootstrap versions are slightly more conservative, but it is still clear that empirical residual tails improve the forecast path materially relative to fixed parametric tails.

5.5.2 Empirical tails with GJR-GARCH

The same empirical-tail idea can then be combined with asymmetric volatility with the empirical residual quantile replacing the Gaussian or Student- t tail.

Model	Viol.	Rate	Kupiec p	Christ. p	CC p	Mean Check Loss
GJR- Gaussian- FHS	8	0.94%	0.8672	0.6961	0.9137	0.0004012
GJR- t -FHS	6	0.71%	0.3664	0.7698	0.6371	0.0004171

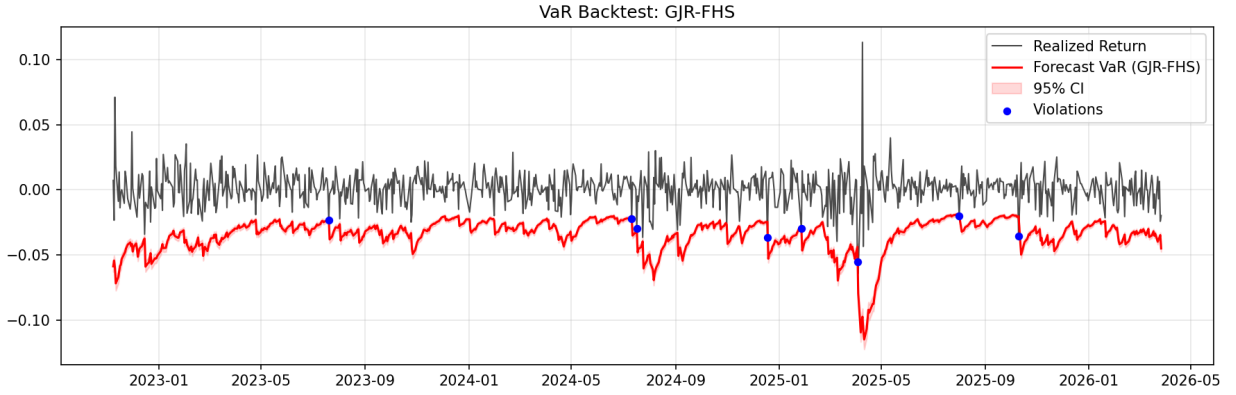


Figure 12: GJR-Gaussian-FHS backtest plot.

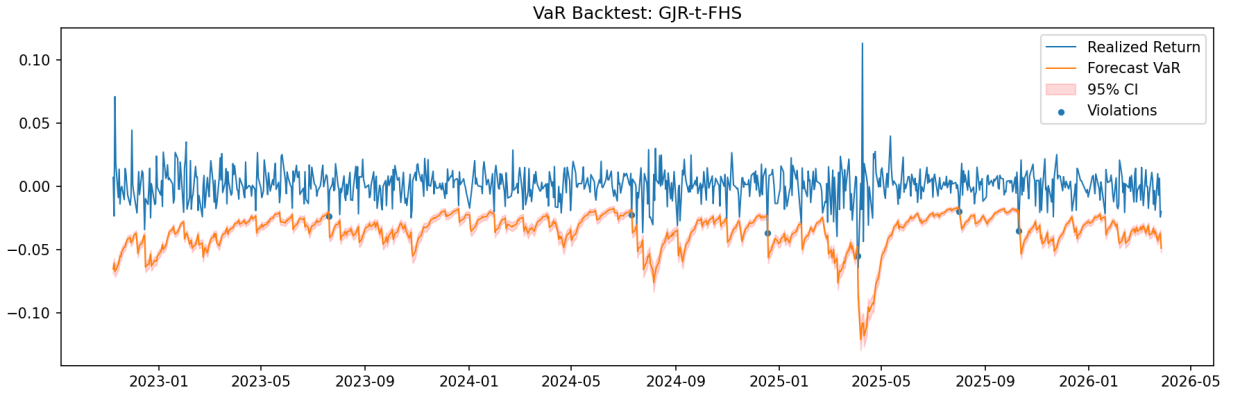


Figure 13: GJR- t -FHS backtest plot.

The paired GJR empirical-tail plots complete the stepwise story. Both are visibly more responsive in the left tail than the earlier GJR-GARCH plot, but GJR-Gaussian-FHS gives the best overall balance between responsiveness and calibration. It deepens quickly in stress without drifting into excessive conservatism afterward. That visual balance is consistent with its strongest overall backtest performance.

5.5.3 Summary of the stepwise comparison

The main methodological lesson is that empirical tail calibration contributes more than further refinements to the variance recursion alone. Student- t innovations improve the fit, and asymmetry provides some additional benefit, but the strongest improvement comes from estimating the lower tail empirically.

6 Forecast Uncertainty and 95% Confidence Intervals

A VaR forecast is an estimate rather than a fixed population quantity. It depends on estimated volatility, estimated or empirical tail quantiles, and finite-sample information. This section quantifies the uncertainty around the point forecast by discussing the logic of quantile confidence intervals and how they are constructed for different models in this study.

6.1 General quantile-CI logic

A common large-sample approximation for a quantile standard error is

$$\text{SE}(\hat{q}_\alpha) = \sqrt{\frac{\alpha(1-\alpha)}{nf(\hat{q}_\alpha)^2}},$$

which leads to the generic 95% interval

$$\hat{q}_\alpha \pm 1.96 \text{ SE}(\hat{q}_\alpha).$$

For location-scale VaR models,

$$\widehat{\text{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{q}_\alpha,$$

so quantile uncertainty translates directly into VaR uncertainty.

6.2 Model-specific CI constructions

The reported `lower_bound` and `upper_bound` values are model-specific rather than generated by one universal formula.

6.2.1 Gaussian parametric models

For Gaussian models such as RiskMetrics, Gaussian GARCH, and GARCHX, the Gaussian quantile standard error is mapped through

$$\hat{\mu} + \hat{\sigma} (\Phi^{-1}(\alpha) \pm 1.96 \text{ SE}(\hat{q}_\alpha)).$$

6.2.2 Student- t parametric models

For Student- t parametric models, the same logic is used with the standardized Student- t density and quantile $\hat{q}_\alpha = t_\nu^{-1}(\alpha)\sqrt{\frac{\nu-2}{\nu}}$ in place of the Gaussian ones.

6.2.3 FHS and empirical residual quantile models

For FHS models, the empirical α -quantile of standardized residuals is first estimated and its sampling uncertainty is then mapped through the one-step volatility forecast.

6.2.4 Bootstrap models

For bootstrap models, standardized residuals are resampled, one-step returns are simulated, and the interval is built from the simulated predictive distribution.

6.2.5 Historical Simulation and CAViaR-SAV

Historical Simulation uses sample-quantile interval logic on recent raw returns. CAViaR-SAV does not produce a direct likelihood-based interval in the current implementation, so its interval is a proxy based on a recentered empirical return sample.

6.3 Interval-width comparison

Model	Mean CI width
CAViaR-SAV	0.02064
Historical Simulation	0.02064
GARCH- t -FHS	0.00783
GARCH-FHS	0.00688
GARCH(1,1) with Student- t innovations	0.00677
GJR-GARCH	0.00637
GJR- t -FHS	0.00586
GJR-FHS	0.00436
GARCH-Gaussian	0.00369
GARCH- t -Bootstrap	0.00368
RiskMetrics ($\lambda = 0.94$)	0.00368
GARCHX(VIX)	0.00353
GARCH-Bootstrap	0.00303

The leading models are not those with the narrowest intervals, but those that combine reasonable interval width with good calibration. Narrow intervals alone are not evidence of better risk forecasting if the VaR level itself is too shallow.

7 Integrated Final Comparison Table

Table 1 summarizes the full model comparison. Short labels are used only here to keep the table compact. They are defined in the note below.

Model	Viol.	Rate	Kupiec p	Christ. p	CC p	Mean Check Loss	Verdict
GJR-FHS	8	0.94%	0.8672	0.6961	0.9137	0.0004012	Best
GARCH-FHS	7	0.83%	0.5984	0.7327	0.8212	0.0004126	Strong
GARCH-t-FHS	6	0.71%	0.3664	0.7698	0.6371	0.0004258	Strong
GJR-t-FHS	6	0.71%	0.3664	0.7698	0.6371	0.0004171	Strong
GARCH-t-Boot	6	0.71%	0.3664	0.7698	0.6371	0.0004302	Strong
GARCH-Boot	5	0.59%	0.1934	0.8075	0.4166	0.0004154	Strong
HS	12	1.42%	0.2528	0.1569	0.1910	0.0004826	Benchmark
GARCH-t	13	1.53%	0.1480	0.1890	0.1482	0.0004131	Mid
CAViaR-SAV	6	0.71%	0.3664	0.0297	0.0625	0.0004010	Mixed
GJR-GARCH	15	1.77%	0.0424	0.4621	0.0972	0.0004069	Mixed
GARCHX(VIX)	15	1.77%	0.0424	0.2621	0.0679	0.0004185	Weak
RM(0.94)	18	2.12%	0.0043	0.3925	0.0117	0.0004458	Weak
GARCH-N	22	2.59%	0.0001	0.5970	0.0005	0.0004363	Weak

Note: HS = Historical Simulation; RM(0.94) = RiskMetrics with ($\lambda = 0.94$); GARCH-N = Gaussian GARCH(1,1); GARCH-t = GARCH(1,1) with Student- t innovations; GARCH-t-FHS and GJR-t-FHS use Student- t fitted volatility with empirical residual tails.

The table reinforces the main narrative. GJR-FHS is the strongest overall model because its violation rate is closest to the nominal 1% target and its backtest p -values are uniformly strong. GARCH-FHS is a close second and is the simplest strong empirical-tail recommendation. Historical Simulation remains a serious benchmark. GARCHX(VIX) is informative as a comparison benchmark but not competitive enough for the final recommendation. CAViaR-SAV remains a mixed result. It is attractive on mean check loss and unconditional coverage, but weakened by dependence in the timing of violations and by the simplified implementation.

8 Discussion

The empirical comparison points to a clear methodological conclusion. For one-day-ahead 1% VaR on QQQ, the hardest part is not volatility modeling by itself, but accurate calibration of the lower tail once volatility has been forecast.

The Gaussian baselines fail first, showing that volatility dynamics alone do not solve the problem. Student- t innovations improve performance, which confirms that tail thickness matters. Richer variance dynamics by themselves help less, GARCHX(VIX) is not persuasive in this setting, and GJR-GARCH helps only modestly until empirical tails are added. Historical Simulation performs well enough to remain a serious benchmark, while the simplified CAViaR-SAV result is interesting but not convincing enough to displace the leading models.

The decisive improvement occurs when standardized residual tails are estimated empirically. That is why the final recommendation is **GJR GARCH-FHS**, with **GARCH-FHS** a close second.

9 Limitations

Several limitations should be stated directly.

1. **GARCHX is not fully justified:** VIX is included as an exogenous regressor, but the project does not formally verify that it adds forecasting value beyond plain GARCH.
2. **The CAViaR-SAV implementation is simplified.** Only the SAV specification is used, optimization is relatively light, and the Dynamic Quantile test is not included.
3. **Bootstrap forecasts are reproducible by construction.** The implementation resets the random seed inside the bootstrap forecast function, which improves reproducibility but makes the rolling bootstrap less genuinely stochastic than a fresh-resampling design would be.
4. **The conclusions are application-specific.** The ranking is for QQQ, a one-day horizon, and a 1% VaR target.

10 Conclusion

This report reorganizes the VaR study as a stepwise model-selection exercise. The starting point was a pair of Gaussian volatility-based VaR methods: RiskMetrics ($\lambda = 0.94$) and Gaussian GARCH(1,1). Both produced too many violations, showing that Gaussian lower tails are too thin for QQQ one-day 1% VaR forecasting.

The next step was GARCH(1,1) with Student- t innovations. This improved performance, confirming that heavier tails matter, but it did not fully solve the problem. The extension path then split in two. In the richer-volatility direction, GARCHX(VIX) was not strong enough to justify itself as the main path, while GJR-GARCH suggested that asymmetry was somewhat useful. In the alternative-method direction, Historical Simulation proved to be a serious benchmark, and the simplified CAViaR-SAV implementation remained interesting but not convincing enough because of dependence failure and implementation limits.

The final and most important improvement came from empirical-tail calibration. Applying FHS and related bootstrap methods to plain GARCH already improved performance materially. Applying

the same idea to GJR-GARCH improved further and produced the best overall final model. The final recommendation is therefore GJR-FHS, with GARCH-FHS a close second.

11 References

- Barone-Adesi, G., Giannopoulos, K., & Vosper, L. (1999). VaR without correlations for portfolios of derivative securities. *Journal of Futures Markets*, 19(5), 583–602.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.
- Christoffersen, P. F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862.
- Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- Engle, R. F., & Manganelli, S. (2004). CAViaR: Conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, 22(4), 367–381.
- J.P. Morgan/Reuters. (1996). *RiskMetrics technical document* (4th ed.).
- Kupiec, P. H. (1995). Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2), 73–82.
- Pascual, L., Romo, J., & Ruiz, E. (2006). Bootstrap prediction for returns and volatilities in GARCH models. *Computational Statistics & Data Analysis*, 50(9), 2293–2312.

12 Appendix. Representative Implementation Snippets

This appendix includes short method-relevant snippets only. For full code, see the project repository.

<https://github.com/eyao0906/One-day-ahead-1-VaR-forecasting-on-QQQ-daily-log-returns>

12.1 RiskMetrics recursion and VaR mapping

```
sigma2 = np.var(r[:init_window])
for i in range(init_window, n):
    sigma2 = lambd * sigma2 + (1.0 - lambd) * (r[i] ** 2)

sigma_next = np.sqrt(max(sigma2, 1e-12))
mu = float(np.mean(r))
var = mu + sigma_next * norm.ppf(alpha)
```

12.2 GARCH one-step VaR with Gaussian / Student-*t* innovations

```
y = returns * 100.0
model = arch_model(y, mean="Constant", vol="GARCH", p=1, q=1, dist=("StudentsT" if dist == "t"
res = model.fit(dispatch="off")

if dist == "t" and "nu" in res.params.index:
    nu = float(res.params["nu"])
    q = float(t.ppf(_safe_quantile(alpha), df=max(nu, 2.2)))
    # standardize t to variance 1
    q *= np.sqrt((nu - 2.0) / nu)
    lower_bound, upper_bound = _parametric_quantile_ci(mu, sigma, alpha, n_eff=len(y), dist=
else:
    nu = np.nan
    q = float(norm.ppf(_safe_quantile(alpha)))
    lower_bound, upper_bound = _parametric_quantile_ci(mu, sigma, alpha, n_eff=len(y), dist=
var = mu + sigma * q
```

12.3 GARCH FHS

```
y = returns * 100.0
# Gaussian: model = arch_model(y, mean="Constant", vol="GARCH", p=1, q=1, dist="normal")
model = arch_model(y, mean="Constant", vol="GARCH", p=1, q=1, dist="StudentsT")
res = model.fit(dispatch="off")

mu = float(res.params.get("mu", 0.0)) / 100.0
sigma2 = float(res.forecast(horizon=1, reindex=False).variance.values[-1, 0]) / (100.0**2)
sigma_next = float(np.sqrt(max(sigma2, 1e-12)))

# Compute standardized residuals and remove initial NaNs
std_resid = res.resid / res.conditional_volatility
std_resid = std_resid[~np.isnan(std_resid)]

# Empirical quantile of the t-standardized residuals
q_emp = float(np.quantile(std_resid, _safe_quantile(alpha)))
q_lo, q_hi = _sample_quantile_ci(std_resid, alpha, point_quantile=q_emp)

var = mu + sigma_next * q_emp
```

12.4 GARCHX

```
y_adj <- r_train - mu
fit_garchx <- garchx(y_adj, order = c(1, 1), xreg = vix_train)
```

```

vix_T <- matrix(tail(vix_train, 1), ncol = 1)
pred_garchx <- predict(fit_garchx, n.ahead = 1, newxreg = vix_T)
sigma_next_x <- sqrt(as.numeric(pred_garchx))
var_garchx <- (mu + sigma_next_x * qnorm(alpha)) / 100

```

12.5 GJR-GARCH

```

y = returns * 100.0
model = arch_model(y, mean="Constant", vol="GARCH", p=1, o=1, q=1, dist=("StudentsT" if dist ==
res = model.fit(disp="off")

mu = float(res.params.get("mu", 0.0)) / 100.0
sigma2 = float(res.forecast(horizon=1, reindex=False).variance.values[-1, 0]) / (100.0**2)
sigma = float(np.sqrt(max(sigma2, 1e-12)))

if dist == "t" and "nu" in res.params.index:
    nu = float(res.params["nu"])
    q = float(t.ppf(_safe_quantile(alpha), df=max(nu, 2.2)))
    q *= np.sqrt((nu - 2.0) / nu)
    lower_bound, upper_bound = _parametric_quantile_ci(mu, sigma, alpha, n_eff=len(y), dist="t")
else:
    nu = np.nan
    q = float(norm.ppf(_safe_quantile(alpha)))
    lower_bound, upper_bound = _parametric_quantile_ci(mu, sigma, alpha, n_eff=len(y), dist="n")
var = mu + sigma * q

```

12.6 GJR-GARCH FHS

```

y = returns * 100.0
model = arch_model(y, mean="Constant", vol="GARCH", p=1, o=1, q=1, dist="normal")
res = model.fit(disp="off")

mu = float(res.params.get("mu", 0.0)) / 100.0
sigma2 = float(res.forecast(horizon=1, reindex=False).variance.values[-1, 0]) / (100.0**2)
sigma_next = float(np.sqrt(max(sigma2, 1e-12)))

std_resid = res.resid / res.conditional_volatility
std_resid = std_resid[~np.isnan(std_resid)]
q_emp = float(np.quantile(std_resid, alpha))
var = mu + sigma_next * q_emp

```

12.7 Historical Simulation

```
tail_sample = np.asarray(returns[-250:], dtype=float)
var = float(np.quantile(tail_sample, alpha))
```

12.8 CAViaR-SAV recursion and check-loss objective

```
@njit(cache=True)
def _caviar_sav_loss(params, r_scaled, alpha):
    b0, b1, b2 = params[0], params[1], params[2]
    v_prev = max(1e-6, -np.quantile(r_scaled, alpha))
    loss = 0.0

    q = -v_prev
    u = r_scaled[0] - q
    loss += u * (alpha - (1.0 if u < 0.0 else 0.0))

    for t in range(1, len(r_scaled)):
        v_t = b0 + b1 * v_prev + b2 * abs(r_scaled[t - 1])
        q_t = -v_t
        u = r_scaled[t] - q_t
        loss += u * (alpha - (1.0 if u < 0.0 else 0.0))
        v_prev = v_t
    return loss

def _caviar_loss_objective(params: np.ndarray, r_scaled: np.ndarray, alpha: float) -> float:
    return float(_caviar_sav_loss(np.asarray(params, dtype=float), r_scaled, float(_safe_quantile(alpha))))

def _fit_caviar_sav(r_scaled: np.ndarray, alpha: float) -> np.ndarray:
    emp_var = max(0.1, -float(np.quantile(r_scaled, _safe_quantile(alpha))))
    starts = [
        np.array([0.05 * emp_var, 0.90, 0.10]),
        np.array([0.10 * emp_var, 0.95, 0.05]),
        np.array([0.20 * emp_var, 0.85, 0.15]),
        np.array([0.01, 0.99, 0.01]),
    ]

    best = None
    for s in starts:
        res = minimize(
            _caviar_loss_objective,
            s,
            args=(r_scaled, alpha),
```



```

        method="Nelder-Mead",
        options={"maxiter": 400, "xatol": 1e-4, "fatol": 1e-4, "disp": False},
    )
    if best is None or res.fun < best.fun:
        best = res
    return np.asarray(best.x, dtype=float)

def _caviar_sav_next_quantile(params: np.ndarray, r_scaled: np.ndarray, alpha: float) -> float:
    b0, b1, b2 = params
    v_prev = max(1e-6, -float(np.quantile(r_scaled, _safe_quantile(alpha))))

    for t in range(1, len(r_scaled)):
        v_prev = b0 + b1 * v_prev + b2 * abs(r_scaled[t - 1])
    v_next = b0 + b1 * v_prev + b2 * abs(r_scaled[-1])
    return -float(v_next)

def forecast_caviar_sav(returns: np.ndarray, alpha: float = 0.01) -> ForecastResult:
    r_scaled = np.asarray(returns, dtype=float) * 100.0
    params = _fit_caviar_sav(r_scaled, alpha)
    q_next_scaled = _caviar_sav_next_quantile(params, r_scaled, alpha)

    proxy_window = min(250, len(r_scaled))
    lower_scaled, upper_scaled = _centered_sample_ci(q_next_scaled, r_scaled[-proxy_window:], alpha)

    return ForecastResult(
        var=q_next_scaled / 100.0,
        meta={
            "b0": float(params[0]),
            "b1": float(params[1]),
            "b2": float(params[2]),
            "var_level": -q_next_scaled / 100.0,
            "lower_bound": float(lower_scaled / 100.0),
            "upper_bound": float(upper_scaled / 100.0),
        },
    )

```

12.9 FHS empirical residual quantile construction

```

std_resid = res.resid / res.conditional_volatility
std_resid = std_resid[~np.isnan(std_resid)]
q_emp = float(np.quantile(std_resid, alpha))
var = mu + sigma_next * q_emp

```

12.10 Bootstrap one-step simulation

```
std_resid = res.resid / res.conditional_volatility
std_resid = std_resid[~np.isnan(std_resid)]

np.random.seed(42)
boot_z = np.random.choice(std_resid, size=n_boot, replace=True)
boot_returns = mu + sigma_next * boot_z
var = float(np.quantile(boot_returns, alpha))
```

12.11 Kupiec unconditional coverage test

```
def kupiec_test(violations: np.ndarray, alpha: float) -> Tuple[float, float]:
    v = np.asarray(violations).astype(int)
    n = len(v)
    x = int(v.sum())
    pi_hat = np.clip(x / max(n, 1), 1e-8, 1 - 1e-8)
    a = np.clip(alpha, 1e-8, 1 - 1e-8)

    lr = -2.0 * (
        (n - x) * np.log(1 - a)
        + x * np.log(a)
        - (n - x) * np.log(1 - pi_hat)
        - x * np.log(pi_hat)
    )
    pval = float(1 - chi2.cdf(lr, df=1))
    return float(lr), pval
```

12.12 Christoffersen Independence Test

```
def christoffersen_independence_test(violations: np.ndarray) -> Tuple[float, float]:
    v = np.asarray(violations).astype(int)
    if len(v) < 2:
        return float("nan"), float("nan")

    n00 = n01 = n10 = n11 = 0
    for i in range(1, len(v)):
        prev, cur = v[i - 1], v[i]
        if prev == 0 and cur == 0:
            n00 += 1
        elif prev == 0 and cur == 1:
            n01 += 1
```

```

elif prev == 1 and cur == 0:
    n10 += 1
else:
    n11 += 1

p01 = n01 / max(n00 + n01, 1)
p11 = n11 / max(n10 + n11, 1)
p = (n01 + n11) / max(n00 + n01 + n10 + n11, 1)

p01 = np.clip(p01, 1e-8, 1 - 1e-8)
p11 = np.clip(p11, 1e-8, 1 - 1e-8)
p = np.clip(p, 1e-8, 1 - 1e-8)

ll_ind = (n00 + n10) * np.log(1 - p) + (n01 + n11) * np.log(p)
ll_dep = n00 * np.log(1 - p01) + n01 * np.log(p01) + n10 * np.log(1 - p11) + n11 * np.log(p11)

lr = -2.0 * (ll_ind - ll_dep)
pval = float(1 - chi2.cdf(lr, df=1))
return float(lr), pval

```

12.13 Check-loss calculation

```

u = y - q
check_loss = u * (alpha - (u < 0).astype(float))
mean_check_loss = float(np.mean(check_loss))

```

12.14 Backtesting logic

```

violation = int(realized < var)
lr_uc, p_uc = kupiec_test(v, alpha)
lr_ind, p_ind = christoffersen_independence_test(v)
lr_cc = lr_uc + lr_ind

```